

Q 12

head = ① → ② → ③ → ④ → Null

We have to swap in pairs.

i.e new head = ② → ① → ④ → ③ → Null.

Step 1:- Sabse pehle hm ek dummy node create karenge aur usme 0 store karenge, aur dummy ke next me head ko point kar denge.

i.e.

```
{ ListNode dummy = new ListNode(0);  
  dummy.next = head; }
```

uske baad hm dummy ko prev me store karenge.

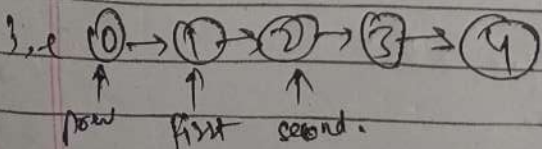
```
i.e. { ListNode prev = dummy; }
```

Step 2:- Phir ek loop chalayenge ki.

```
while (prev.next != null && prev.next.next != null) {
```

```
  ListNode first = prev.next; // matlab first me  
                               hmlog prev.next ko  
                               store karenge.
```

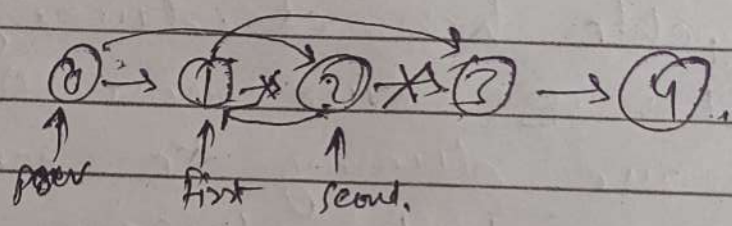
```
  ListNode second = prev.nextfirst.next; // matlab second me hmlog  
                               first.next or prev.  
                               next.next ko store  
                               karenge.
```



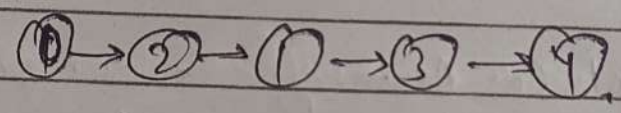
step 3 - // swap :-

$first_next = second_next;$ // break
 $second_next = first;$ // reverse
 $prev_next = second;$ // connect

i.e

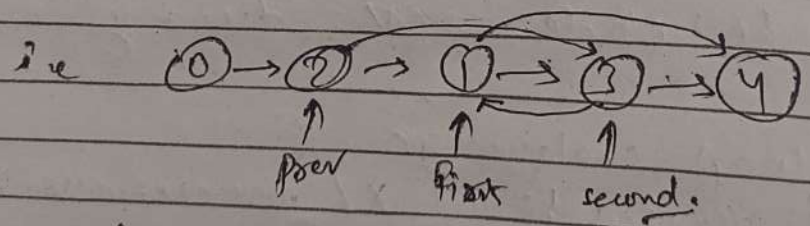


ie.



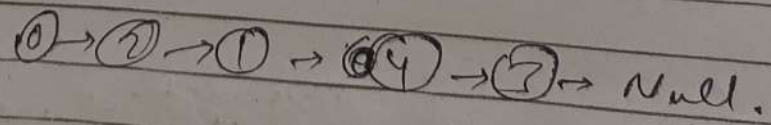
Ab. pointer move karega.

$prev = first;$



* again swap karega.

Finally



At the end

return dummynext;